

# lab7

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## Q1

- code

```
--Q1
SELECT location_id,
       street_address,
       city,
       state_province,
       country_name AS country
FROM locations
NATURAL JOIN countries;
```

- output

|   | LOCATION_ID | STREET_ADDRESS                           | CITY                | STATE_PROVINCE | COUNTRY                  |
|---|-------------|------------------------------------------|---------------------|----------------|--------------------------|
| 1 | 1800        | 460 Bloor St. W.                         | Toronto             | Ontario        | Canada                   |
| 2 | 2500        | Magdalen Centre, The Oxford Science Park | Oxford              | Oxford         | United Kingdom           |
| 3 | 1500        | 2011 Interiors Blvd                      | South San Francisco | California     | United States of America |
| 4 | 1400        | 2014 Jabberwocky Rd                      | Southlake           | Texas          | United States of America |
| 5 | 1700        | 2004 Charade Rd                          | Seattle             | Washington     | United States of America |

## Q2

- code

```
--Q2
SELECT e.last_name,
       e.department_id,
       d.department_name
FROM employees e
JOIN departments d
ON e.department_id = d.department_id;
```

- output

|    | 切换到图表     | DEPARTMENT_ID | DEPARTMENT_NAME |
|----|-----------|---------------|-----------------|
| 1  | Whalen    | 10            | Administration  |
| 2  | Hartstein | 20            | Marketing       |
| 3  | Fay       | 20            | Marketing       |
| 4  | Mourgos   | 50            | Shipping        |
| 5  | Vargas    | 50            | Shipping        |
| 6  | Matos     | 50            | Shipping        |
| 7  | Davies    | 50            | Shipping        |
| 8  | Rajs      | 50            | Shipping        |
| 9  | Lorentz   | 60            | IT              |
| 10 | Ernst     | 60            | IT              |
| 11 | Hunold    | 60            | IT              |
| 12 | Abel      | 80            | Sales           |
| 13 | Zlotkey   | 80            | Sales           |
| 14 | Taylor    | 80            | Sales           |
| 15 | De Haan   | 90            | Executive       |
| 16 | King      | 90            | Executive       |
| 17 | Kochhar   | 90            | Executive       |
| 18 | Gietz     | 110           | Accounting      |
| 19 | Higgins   | 110           | Accounting      |

## Q3

- code

```
--Q3
SELECT e.last_name,
       e.job_id,
       e.department_id,
       d.department_name
FROM employees e
JOIN departments d
ON e.department_id = d.department_id
JOIN locations l
ON d.location_id = l.location_id
WHERE l.city = 'Toronto';
```

- output

|   | LAST_NAME | JOB_ID | DEPARTMENT_ID | DEPARTMENT_NAME |
|---|-----------|--------|---------------|-----------------|
| 1 | Hartstein | MK_MAN | 20            | Marketing       |
| 2 | Fay       | MK_REP | 20            | Marketing       |

## Q4

- code

```
--Q4
SELECT e.last_name AS employee,
       e.employee_id AS emp#,
       m.last_name AS manager,
       m.employee_id AS mgr#
FROM employees e
JOIN employees m
ON e.manager_id = m.employee_id;
```

- output

|    | EMPLOYEE | EMP# | MANAGER   | MGR# |
|----|----------|------|-----------|------|
| 5  | Mourgos  | 124  | King      | 100  |
| 6  | Higgins  | 205  | Kochhar   | 101  |
| 7  | Whalen   | 200  | Kochhar   | 101  |
| 8  | Hunold   | 103  | De Haan   | 102  |
| 9  | Ernst    | 104  | Hunold    | 103  |
| 10 | Lorentz  | 107  | Hunold    | 103  |
| 11 | Davies   | 142  | Mourgos   | 124  |
| 12 | Matos    | 143  | Mourgos   | 124  |
| 13 | Vargas   | 144  | Mourgos   | 124  |
| 14 | Rajs     | 141  | Mourgos   | 124  |
| 15 | Grant    | 178  | Zlotkey   | 149  |
| 16 | Abel     | 174  | Zlotkey   | 149  |
| 17 | Taylor   | 176  | Zlotkey   | 149  |
| 18 | Fay      | 202  | Hartstein | 201  |
| 19 | Gietz    | 206  | Higgins   | 205  |

## Q5

- code

```
--Q5
SELECT e.last_name AS employee,
       e.employee_id AS emp#,
       m.last_name AS manager,
       m.employee_id AS mgr#
FROM employees e
LEFT JOIN employees m
ON e.manager_id = m.employee_id
ORDER BY e.employee_id;
```

- output

|    | EMPLOYEE  | EMP# | MANAGER   | MGR#   |
|----|-----------|------|-----------|--------|
| 1  | King      | 100  | <null>    | <null> |
| 2  | Kochhar   | 101  | King      | 100    |
| 3  | De Haan   | 102  | King      | 100    |
| 4  | Hunold    | 103  | De Haan   | 102    |
| 5  | Ernst     | 104  | Hunold    | 103    |
| 6  | Lorentz   | 107  | Hunold    | 103    |
| 7  | Mourgos   | 124  | King      | 100    |
| 8  | Rajs      | 141  | Mourgos   | 124    |
| 9  | Davies    | 142  | Mourgos   | 124    |
| 10 | Matos     | 143  | Mourgos   | 124    |
| 11 | Vargas    | 144  | Mourgos   | 124    |
| 12 | Zlotkey   | 149  | King      | 100    |
| 13 | Abel      | 174  | Zlotkey   | 149    |
| 14 | Taylor    | 176  | Zlotkey   | 149    |
| 15 | Grant     | 178  | Zlotkey   | 149    |
| 16 | Whalen    | 200  | Kochhar   | 101    |
| 17 | Hartstein | 201  | King      | 100    |
| 18 | Fay       | 202  | Hartstein | 201    |
| 19 | Higgins   | 205  | Kochhar   | 101    |
| 20 | Gietz     | 206  | Higgins   | 205    |

## Q6

- code

```
--Q6
SELECT e1.last_name AS employee,
       e1.department_id,
       e2.last_name AS colleague
FROM employees e1
JOIN employees e2
ON e1.department_id = e2.department_id
AND e1.employee_id <> e2.employee_id
ORDER BY e1.last_name;
```

- output

|    | EMPLOYEE  | DEPARTMENT_ID | COLLEAGUE    |
|----|-----------|---------------|--------------|
| 1  | Abel      |               | 80 Zlotkey   |
| 2  | Abel      |               | 80 Taylor    |
| 3  | Davies    |               | 50 Vargas    |
| 4  | Davies    |               | 50 Mourgog   |
| 5  | Davies    |               | 50 Rajs      |
| 6  | Davies    |               | 50 Matos     |
| 7  | De Haan   |               | 90 Kochhar   |
| 8  | De Haan   |               | 90 King      |
| 9  | Ernst     |               | 60 Hunold    |
| 10 | Ernst     |               | 60 Lorentz   |
| 11 | Fay       |               | 20 Hartstein |
| 12 | Gietz     |               | 110 Higgins  |
| 13 | Hartstein |               | 20 Fay       |
| 14 | Higgins   |               | 110 Gietz    |
| 15 | Hunold    |               | 60 Lorentz   |
| 16 | Hunold    |               | 60 Ernst     |
| 17 | King      |               | 90 Kochhar   |
| 18 | King      |               | 90 De Haan   |
| 19 | Kochhar   |               | 90 King      |
| 20 | Kochhar   |               | 90 De Haan   |
| 21 | Lorentz   |               | 60 Ernst     |
| 22 | Lorentz   |               | 60 Hunold    |
| 23 | Matos     |               | 50 Rajs      |
| 24 | Matos     |               | 50 Vargas    |
| 25 | Matos     |               | 50 Davies    |
| 26 | Matos     |               | 50 Mourgog   |

## Q7

- code

```
--Q7
DESC job_grades;

SELECT e.last_name,
       e.job_id,
       d.department_name,
       e.salary,
       g.grade_level
FROM employees e
JOIN departments d
ON e.department_id = d.department_id
JOIN job_grades g
ON e.salary BETWEEN g.lowest_sal AND g.highest_sal;
```

- output

|    | LAST_NAME | JOB_ID     | DEPARTMENT_NAME      | SALARY   | GRADE_LEVEL |
|----|-----------|------------|----------------------|----------|-------------|
| 1  | King      | AD_PRES    | JOB_ID: VARCHAR2(10) | 24000.00 | E           |
| 2  | De Haan   | AD_VP      | Executive            | 17000.00 | E           |
| 3  | Kochhar   | AD_VP      | Executive            | 17000.00 | E           |
| 4  | Hartstein | MK_MAN     | Marketing            | 13000.00 | D           |
| 5  | Higgins   | AC_MGR     | Accounting           | 12000.00 | D           |
| 6  | Abel      | SA_REP     | Sales                | 11000.00 | D           |
| 7  | Zlotkey   | SA_MAN     | Sales                | 10500.00 | D           |
| 8  | Hunold    | IT_PROG    | IT                   | 9000.00  | C           |
| 9  | Taylor    | SA_REP     | Sales                | 8600.00  | C           |
| 10 | Gietz     | AC_ACCOUNT | Accounting           | 8300.00  | C           |
| 11 | Ernst     | IT_PROG    | IT                   | 6000.00  | C           |
| 12 | Fay       | MK_REP     | Marketing            | 6000.00  | C           |
| 13 | Mourgos   | ST_MAN     | Shipping             | 5800.00  | B           |
| 14 | Whalen    | AD_ASST    | Administration       | 4400.00  | B           |
| 15 | Lorentz   | IT_PROG    | IT                   | 4200.00  | B           |
| 16 | Rajs      | ST_CLERK   | Shipping             | 3500.00  | B           |
| 17 | Davies    | ST_CLERK   | Shipping             | 3100.00  | B           |
| 18 | Matos     | ST_CLERK   | Shipping             | 2600.00  | A           |
| 19 | Vargas    | ST_CLERK   | Shipping             | 2500.00  | A           |

## Q8

- code

```
--Q8
SELECT e.last_name,
       e.hire_date
FROM employees e
WHERE e.hire_date >
      (SELECT hire_date
       FROM employees
       WHERE last_name = 'Davies');
```

- output

|   | LAST_NAME | HIRE_DATE  |
|---|-----------|------------|
| 1 | Lorentz   | 1999-02-07 |
| 2 | Mourgos   | 1999-11-16 |
| 3 | Matos     | 1998-03-15 |
| 4 | Vargas    | 1998-07-09 |
| 5 | Zlotkey   | 2000-01-29 |
| 6 | Taylor    | 1998-03-24 |
| 7 | Grant     | 1999-05-24 |
| 8 | Fay       | 1997-08-17 |

## Q9

- code

```
--Q9
SELECT e.last_name AS employee,
       e.hire_date AS emp_hire_date,
       m.last_name AS manager,
       m.hire_date AS mgr_hire_date
```

```
FROM employees e
JOIN employees m
ON e.manager_id = m.employee_id
WHERE e.hire_date < m.hire_date;
```

- output

|   | EMPLOYEE | EMP_HIRE_DATE | MANAGER | MGR_HIRE_DATE |
|---|----------|---------------|---------|---------------|
| 1 | Whalen   | 1987-09-17    | Kochhar | 1989-09-21    |
| 2 | Hunold   | 1990-01-03    | De Haan | 1993-01-13    |
| 3 | Davies   | 1997-01-29    | Mourgos | 1999-11-16    |
| 4 | Matos    | 1998-03-15    | Mourgos | 1999-11-16    |
| 5 | Vargas   | 1998-07-09    | Mourgos | 1999-11-16    |
| 6 | Rajs     | 1995-10-17    | Mourgos | 1999-11-16    |
| 7 | Grant    | 1999-05-24    | Zlotkey | 2000-01-29    |
| 8 | Abel     | 1996-05-11    | Zlotkey | 2000-01-29    |
| 9 | Taylor   | 1998-03-24    | Zlotkey | 2000-01-29    |

## Q10

- code

```
--Q10

SELECT department_id
FROM departments
MINUS
SELECT DISTINCT department_id
FROM employees
WHERE job_id = 'ST_CLERK';
```

- output

|   | DEPARTMENT_ID |  |
|---|---------------|--|
| 1 | 10            |  |
| 2 | 20            |  |
| 3 | 60            |  |
| 4 | 80            |  |
| 5 | 90            |  |
| 6 | 110           |  |
| 7 | 190           |  |

## Q11

- code

```
--Q11
SELECT e.employee_id,
       e.job_id
FROM employees e
JOIN job_history j
ON e.employee_id = j.employee_id
WHERE e.job_id = j.job_id;
```

- output

|   | EMPLOYEE_ID | JOB_ID  |
|---|-------------|---------|
| 1 | 176         | SA_REP  |
| 2 | 200         | AD_ASST |